RADU ECHIM

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EDUCATION

PhD in AI & Robotics
University College London
Sep 2023 - Present | London, UK
- Perceptive robot manipulation learning using generative models.

CUDA Course for GPU Programming
University of Oxford
July 2024 | Oxford, UK
- Learned basic and advanced concepts about NVIDIA's CUDA platform. Implemented ML architectures with a focus on memory optimization and performance tuning.

MEng Computer Science
University College London
Sep 2019 – Jun 2023 | London, UK
- Received First Honours distinction
- Minor in Applied Mathematics and Machine Learning.
- Award for **best research project** in 2020. Designed ML algorithm for autonomous delivery and efficient matching algorithms using supervised and unsupervised learning methods.
- Award for Outstanding IBM x UCL Project in 2021. Researched the architectural design and developed a low-latency, low resource distributed system for video conferencing.

EXPERIENCE

Software Engineer
@ S&P Global
Sep 2021 - Aug 2022 | London UK & Remote
- Portfolio Valuations team within Market Intelligence department, responsible for developing large-scale market simulation and implementing numerical methods.
- Created additional pricing capabilities to OTC products with financial analysts. Increased performance and scalability of valuation algorithms.
- Stress-testing portfolios and risk sensitivities calculations to ensure regulatory compliance of Initial Margin across client portfolios.

Software Engineering Intern
@ Tellence Technologies
Sep - Dec 2020 | Bucharest, Romania
- Software engineering team responsible for LiveU's outsourced projects. LiveU is a provider of 5G live streaming capabilities for CNN and FoxNews.
- Created new API endpoints on the back-end for the unit simulator using Java and Spring to improve stress testing capabilities between servers and web application.

Programming Tutor
@ University College London
Sep 2020 - May 2021 | London, UK
- Taught CS & ML algorithms and data structures in Python, C/C++ using Torch.

SKILLS

Machine Learning | Robotics | Python | Torch | C/C++ | CUDA | GPU Programming | Git | Algorithms & Data Structures

PROJECTS

VLA Finetuning for Static Manipulation
January 2025 - Present
- Finetuning large, pretrained vision-language-action models (e.g., π_0) on fine-manipulation insertion tasks.
- Efficient data generation in simulation using our motion planning framework. Created data augmentation pipeline using synthetic failure cases to improve policy performance.

Robotic Manipulation with 3D Perception
June 2024 - January 2025
- Working on diffusion-based models to learn robotic manipulation skills such as picking and insertion from small data samples.
- Analyzed generalizability of state-of-the-art models such as 3D Diffusion Policy.
- Identified source of model limitations by interpreting the learned representations related to 3D perception.
- Increased out-of-distribution performance from 0% to 99.5% using 6D pose estimation which removes the limitations of 3D perception.

Parallel Data Generation with Motion Planning
November 2023 - Present
- Framework for generating synthetic robot datasets to be used for supervised learning using classic motion-planning algorithms. Can generate hundreds of realistically-rendered robot trajectories efficiently in parallel.
- Improved policy learning by adding variation to trajectory generation and task randomization.

Personalized Song Generation @ Manele.io
Aug 2025
- Text-to-song generation model for niche music genres an application which reached 20k+ active users and 30k ARR.

Data Valuation for Distributed AI
Jun 2022 - Sep 2023
- Developed algorithm for accurately predicting the task-specific utility of a dataset (data valuation) for an AI model. Based on Neural Tangent Kernel theory.